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V300 ZEFIRO

A New Sense of Speed

CCTV
SFRS

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00	20.09.2011	First issue	
01	27/01/2020	Issue for alignment with CCTV-VF.CONCEPT, baseline 1.3 as built	

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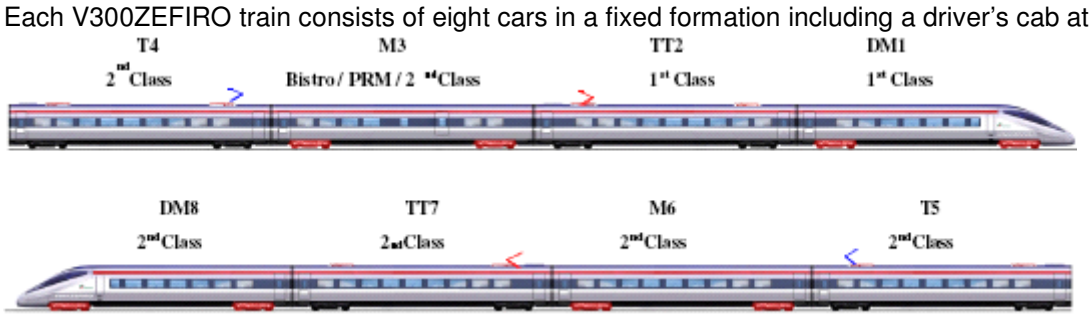
5 References ~~1712~~

1 Introduction

This document is a function specification for the control of the CCTV system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
CCTV	Closed Circuit TeleVision
DVR	Digital Video Recorder
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
LCU	Local Control Unit
MAR	Mobile Access Router
MMI	Man Machine Interface
PIS	Passenger Information System
SCU	System Control Unit
TCMS	Train Control and Management System

2 System Overview

The video surveillance system is completely integrated into the PIS system.
There will be one camera for each vestibule and two cameras for each passenger compartment for end cars and standard cars.
The M3 car will have one camera for the passenger compartment, one for each vestibule and one for the bar bistrò area.
Therefore there will be four cameras for the intermediate cars while there will be three cameras for the end cars.
The video streams are recorded to guarantee 100 hours of recording. The download of information can be anyway performed from a centralized location though the PIS network.
The option for the admission on the Swiss railway network will include two external cameras for each cab to allow the rear view mirror function for each side of the train. In this case two additional LCD monitors will be installed inside the walls of each cab, one for each side to be monitored. Cameras and monitors can be activated automatically with train at zero speed or can be activated manually eg. from drivers display.

2.1 System Configuration

The CCTV system is integrated together with PIS system.
The following figure is a scheme of the system architecture.

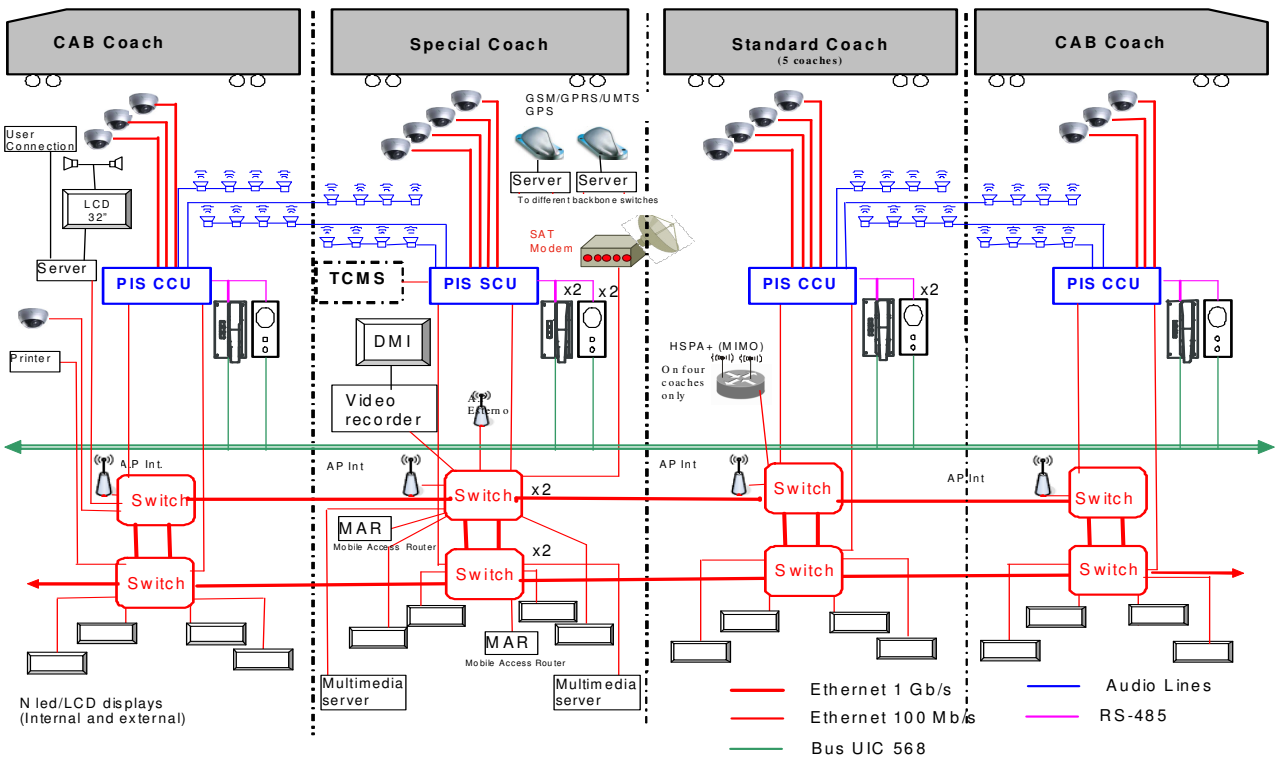


Figure 1: PIS and CCTV architecture

The cameras distribution in the train is according to the following figure:

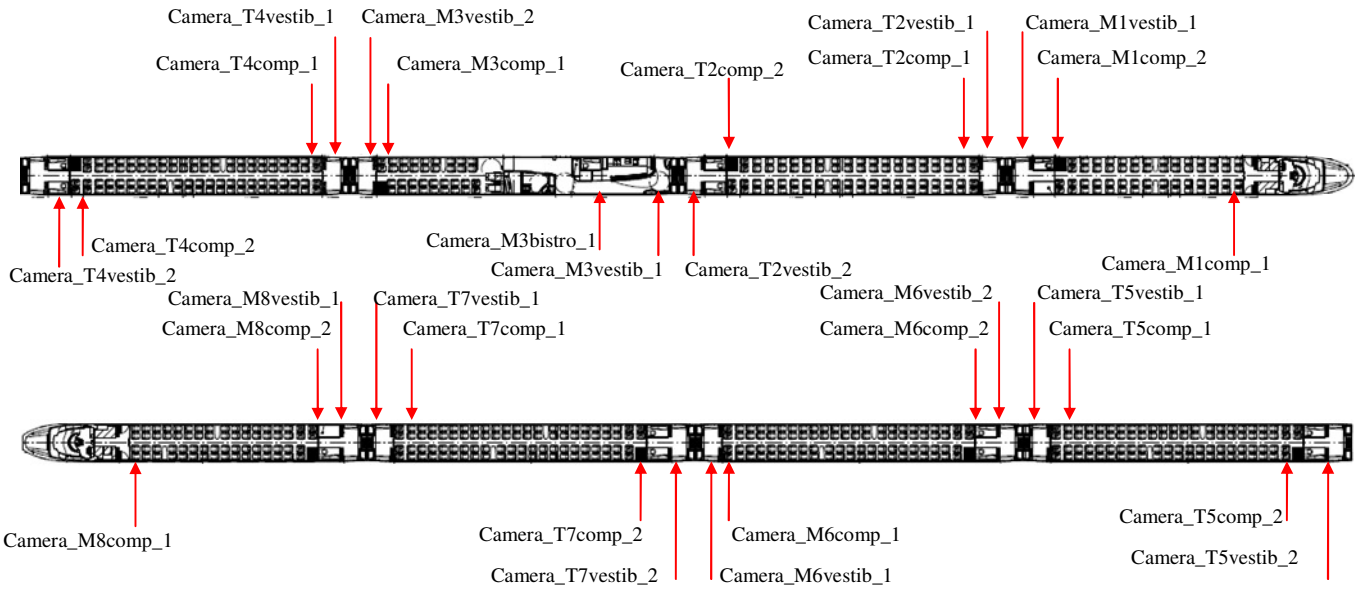


Figure 2: Cameras layout

3 Functional requirements

3.1 General CCTV Control

3.1.1 Architecture

3.1.1.1 Function design

The CCTV is completely integrated into the PIS system. There is no TCMS interaction for the video recording functions. All the information are available in the PIS system.
Information that have to be recorded together with images (TrainNumber, DateTime, and veihcle number) are informationi already available to the PIS/CCTV system. The train number is received through GSM by the ground (OBoE functionality). The date Time is compuded from GPS signal. Veichle number is knonw by the PIS.
In case of degraded situation (loss of ground communication) the train number can be input through the PIS MMI.

3.1.1.1.1 Normal condition

Train number and datetime information is received by the OBoE through GSM and GPS and sent to the DVR (Digital Video Recorder) for image tagging.

3.1.1.1.2 Failure cases

A. failure of GPS => time recorded with images could be not correct
B. failure of GSM connection => train number to be registered with images could be not available
C. failure of PIS network => train number and data time to be registered with images could be not available

3.1.1.1.3 Failure detection[IF1]

A. Failure detected by the PIS and communicated to TCMS
B. Failure detected by the PIS and communicated to TCMS
C. Failure detected by the PIS and communicated to TCMS

3.1.1.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. PIS shall warn the train staff that can manually insert train number. Failure isolated and repaired during maintenance
C. Failure isolated and repaired during maintenance

3.1.1.2 Requirements

ID	2F_13.Zefiro-Europe.TRS.40
Description	Each image shall be recorded together with the information such as: * Train number * Date and time * Vehicle number
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL	China FVDS

2F_13.Zefiro-Europe.CONCEPT.55	CCTV shall use TrainNumber, Date Time, and veihcle number when storing CCTV images	Derived Requirement	4S_09.05.-.Video Surveillance Systems and Closed Circuit Televisions		N/A	

3.1.1.3 Option for the admission on the Swiss railway network

3.1.1.3.1 Function design

The cab will be equipped with two cameras and two displays (on each side of the cab) to act as rearview mirror

3.1.1.3.1.1 Normal condition

When the train is at standstill the two display will show the images of the two cameras. As soon as the train starts the display are switched off

3.1.1.3.1.2 Failure cases

A. TCMS fails to command the display to be switched off => display on while the train is driving

3.1.1.3.1.3 Failure detection

A. Failure detected by the driver

3.1.1.3.1.4 Failure handling

A. Failure isolated and repaired during maintenance

3.1.1.3.2 Requirements

ID	2F_13.Zefiro-Europe.TRS.23
Description	Deleted
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_13.Zefiro-Europe.CONCEPT.60	The TCMS shall command the PIS to switch on the displays in the active cab when the train speed is 0 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_13.Zefiro-Europe.CONCEPT.101	The PIS shall turn on the CCTV displays in the active can when commanded by the TCMS	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A

3.1.2 PIS MMI in staff area

3.1.2.1 Function design

The camera images are normally displayed on the PIS MMI. In case of fire alarm detected onboard the Staff MMI shall visualize the image of the nearest camera to the location of where the fire has been detected.
In case of emergency intercom activation the MMI shall visualize the image of the camera nearer to the Intercom.
This function is completely realized by the PIS/CCTV system.

3.1.2.1.1 Normal condition

When fire is detected onboard the Staff MMI visualizes the image of the nearest camera to the location of where the fire has been detected

3.1.2.1.2 Failure cases

A. TCMS fails to command PIS MMI to display the camera near the fire alarm

3.1.2.1.3 Failure detection

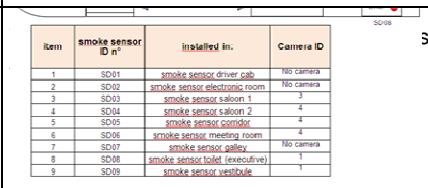
A. Failure not detected

3.1.2.1.4 Failure handling

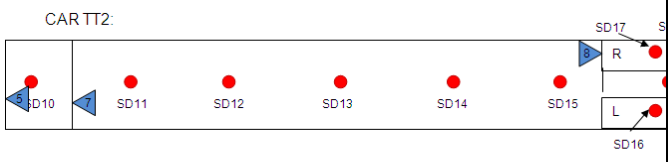
A. Failure not detected

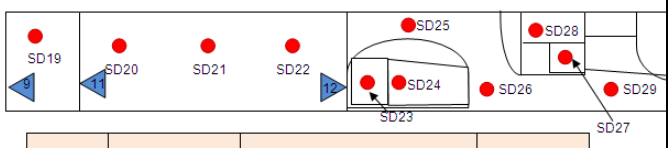
3.1.2.2 Requirements

ID	2F_13.Zefiro-Europe.TRS.37
Description	In case of fire alarm detected onboard the Staff MMI shall visualize the image of the nearest camera to the location of where the fire has been detected.
Safety level	N/A

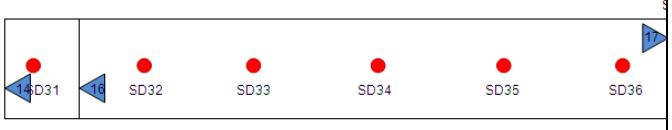
Nr	Description	Type	Derived to	User Interface	SIL
2F_13.Zefiro-Europe.CONCEPT.65	In case of fire alarm the TCMS shall send to PIS SCU indication of the camera nearest to the detection of fire, according below concepts. In case of multiple alarm, TCMS shall send the indication of camera in sequence.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_13.Zefiro-Europe.CONCEPT.146	 shall be activated according	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

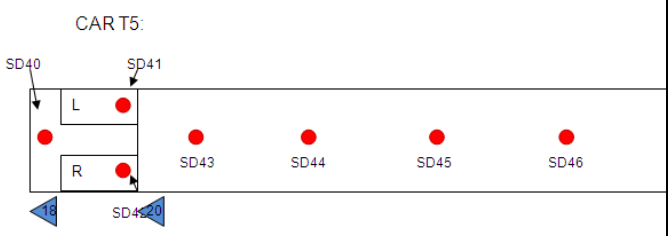
Item	smoke sensor ID n°	installed in.	Camera ID
1	SD-01	smoke sensor driver cab	No camera
2	SD-02	smoke sensor electronic room	No camera
3	SD-03	smoke sensor saloon 1	3
4	SD-04	smoke sensor saloon 2	4
5	SD-05	smoke sensor corridor	4
6	SD-06	smoke sensor meeting room	4
7	SD-07	smoke sensor galley	No camera
8	SD-08	smoke sensor toilet (executive)	1
9	SD-09	smoke sensor vestibule	1

<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL</th></tr>
<tr><td>2F_13.Zefiro-Europe.CONCEPT.147</td><td>When a fire alarm is set in TT2 car, a cam shall be activated according following relation:
CAR TT2:
 

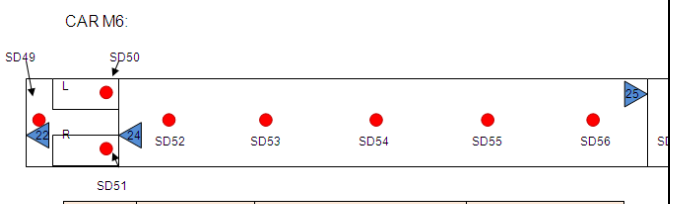
<table>
<tr><th>item</th><th>smoke sensor ID n°</th><th>installed in:</th><th>Camera ID</th></tr>
<tr><td>10</td><td>SD10</td><td>smoke sensor short vestibule</td><td>5</td></tr>
<tr><td>11</td><td>SD11</td><td>smoke sensor saloon 1</td><td>7</td></tr>
<tr><td>12</td><td>SD12</td><td>smoke sensor saloon 2</td><td>7</td></tr>
<tr><td>13</td><td>SD13</td><td>smoke sensor saloon 3</td><td>7</td></tr>
<tr><td>14</td><td>SD14</td><td>smoke sensor saloon 4</td><td>8</td></tr>
<tr><td>15</td><td>SD15</td><td>smoke sensor saloon 5</td><td>8</td></tr>
<tr><td>16</td><td>SD16</td><td>smoke sensor toilet 1 (left)</td><td>6</td></tr>
<tr><td>17</td><td>SD17</td><td>smoke sensor toilet 2 (right)</td><td>6</td></tr>
<tr><td>18</td><td>SD18</td><td>smoke sensor vestibule</td><td>6</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>N/A</td></tr>
<tr><td>2F_13.Zefiro-Europe.CONCEPT.148</td><td>When a fire alarm is set in M3 car, a cam shall be activated according following relation:
CAR M3:
 

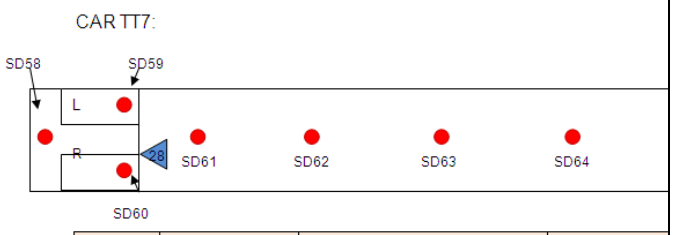
<table>
<tr><th>item</th><th>smoke sensor ID n°</th><th>installed in:</th><th>Camera ID</th></tr>
<tr><td>19</td><td>SD19</td><td>smoke sensor short vestibule</td><td>9</td></tr>
<tr><td>20</td><td>SD20</td><td>smoke sensor saloon 1</td><td>12</td></tr>
<tr><td>21</td><td>SD21</td><td>smoke sensor saloon 2</td><td>11</td></tr>
<tr><td>22</td><td>SD22</td><td>smoke sensor saloon 3</td><td>11</td></tr>
<tr><td>23</td><td>SD23</td><td>smoke sensor train staff</td><td>No camera</td></tr>
<tr><td>24</td><td>SD24</td><td>smoke sensor toilet HK</td><td>No camera</td></tr>
<tr><td>25</td><td>SD25</td><td>smoke sensor corridor 1</td><td>No camera</td></tr>
<tr><td>26</td><td>SD26</td><td>smoke sensor corridor 2</td><td>No camera</td></tr>
<tr><td>27</td><td>SD27</td><td>smoke sensor crew room</td><td>No camera</td></tr>
<tr><td>28</td><td>SD28</td><td>smoke sensor crew toilet</td><td>No camera</td></tr>
<tr><td>29</td><td>SD29</td><td>smoke sensor bistrot 1</td><td>10</td></tr>
<tr><td>30</td><td>SD30</td><td>smoke sensor bistrot 2</td><td>10</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>N/A</td></tr>
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<tr><td>2F_13.Zefiro-Europe.CONCEPT.149</td><td>When a fire alarm is set in T4 car, a cam shall be activated according following relation:
CAR T4:
 

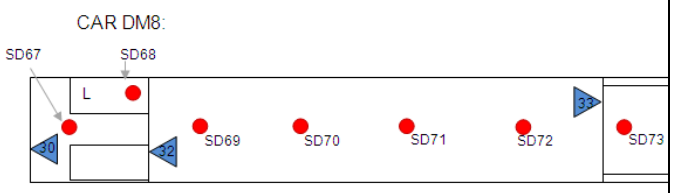
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<tr><th>item</th><th>smoke sensor ID n°</th><th>installed in:</th><th>Camera ID</th></tr>
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<tr><td>32</td><td>SD32</td><td>smoke sensor saloon 1</td><td>16</td></tr>
<tr><td>33</td><td>SD33</td><td>smoke sensor saloon 2</td><td>16</td></tr>
<tr><td>34</td><td>SD34</td><td>smoke sensor saloon 3</td><td>16</td></tr>
<tr><td>35</td><td>SD35</td><td>smoke sensor saloon 4</td><td>17</td></tr>
<tr><td>36</td><td>SD36</td><td>smoke sensor saloon 5</td><td>17</td></tr>
<tr><td>37</td><td>SD37</td><td>smoke sensor toilet 1 (left)</td><td>15</td></tr>
<tr><td>38</td><td>SD38</td><td>smoke sensor toilet 2 (right)</td><td>15</td></tr>
<tr><td>39</td><td>SD39</td><td>smoke sensor vestibule</td><td>15</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>N/A</td></tr>
<tr><td>2F_13.Zefiro-Europe.CONCEPT.150</td><td>When a fire alarm is set in T5 car, a cam shall be activated according following relation:
CAR T5:
 

<table>
<tr><th>item</th><th>smoke sensor ID n°</th><th>installed in:</th><th>Camera I</th></tr>
<tr><td>40</td><td>SD40</td><td>smoke sensor vestibule</td><td>18</td></tr>
<tr><td>41</td><td>SD41</td><td>smoke sensor toilet 1 (left)</td><td>18</td></tr>
<tr><td>42</td><td>SD42</td><td>smoke sensor toilet 2 (right)</td><td>18</td></tr>
<tr><td>43</td><td>SD43</td><td>smoke sensor saloon 5</td><td>20</td></tr>
<tr><td>44</td><td>SD44</td><td>smoke sensor saloon 4</td><td>20</td></tr>
<tr><td>45</td><td>SD45</td><td>smoke sensor saloon 3</td><td>20</td></tr>
<tr><td>46</td><td>SD46</td><td>smoke sensor saloon 2</td><td>21</td></tr>
<tr><td>47</td><td>SD47</td><td>smoke sensor saloon 1</td><td>21</td></tr>
<tr><td>48</td><td>SD48</td><td>smoke sensor short vestibule</td><td>19</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>N/A</td></tr>
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<tr><td>2F_13.Zefiro-Europe.CONCEPT.151</td><td>When a fire alarm is set in M6 car, a cam shall be activated according following relation:
 

<table>
<tr><th>item</th><th>smoke sensor ID n°</th><th>installed in:</th><th>Camera ID</th></tr>
<tr><td>49</td><td>SD49</td><td>smoke sensor vestibule</td><td>22</td></tr>
<tr><td>50</td><td>SD50</td><td>smoke sensor toilet 1 (left)</td><td>22</td></tr>
<tr><td>51</td><td>SD51</td><td>smoke sensor toilet 2 (right)</td><td>22</td></tr>
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<tr><td>53</td><td>SD53</td><td>smoke sensor saloon 4</td><td>24</td></tr>
<tr><td>54</td><td>SD54</td><td>smoke sensor saloon 3</td><td>24</td></tr>
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<tr><td>56</td><td>SD56</td><td>smoke sensor saloon 1</td><td>25</td></tr>
<tr><td>57</td><td>SD57</td><td>smoke sensor short vestibule</td><td>23</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>N/A</td></tr>
<tr><td>2F_13.Zefiro-Europe.CONCEPT.152</td><td>When a fire alarm is set in TT7 car, a cam shall be activated according following relation:
 

<table>
<tr><th>item</th><th>smoke sensor ID n°</th><th>installed in:</th><th>Camera</th></tr>
<tr><td>58</td><td>SD58</td><td>smoke sensor vestibule</td><td>26</td></tr>
<tr><td>59</td><td>SD59</td><td>smoke sensor toilet 1 (left)</td><td>26</td></tr>
<tr><td>60</td><td>SD60</td><td>smoke sensor toilet 2 (right)</td><td>26</td></tr>
<tr><td>61</td><td>SD61</td><td>smoke sensor saloon 5</td><td>28</td></tr>
<tr><td>62</td><td>SD62</td><td>smoke sensor saloon 4</td><td>28</td></tr>
<tr><td>63</td><td>SD63</td><td>smoke sensor saloon 3</td><td>28</td></tr>
<tr><td>64</td><td>SD64</td><td>smoke sensor saloon 2</td><td>29</td></tr>
<tr><td>65</td><td>SD65</td><td>smoke sensor saloon 1</td><td>29</td></tr>
<tr><td>66</td><td>SD66</td><td>smoke sensor short vestibule</td><td>27</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>N/A</td></tr>
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<tr><td>2F_13.Zefiro-Europe.CONCEPT.153</td><td>When a fire alarm is set in DM8 car, a cam shall be activated according following relation:
 

<table><tr><th>item</th><th>smoke sensor ID n°</th><th>installed in:</th><th>Camer</th></tr><tr><td>67</td><td>SD67</td><td>smoke sensor vestibule</td><td>30</td></tr><tr><td>68</td><td>SD68</td><td>smoke sensor toilet</td><td>30</td></tr><tr><td>69</td><td>SD69</td><td>smoke sensor saloon 4</td><td>32</td></tr><tr><td>70</td><td>SD70</td><td>smoke sensor saloon 3</td><td>32</td></tr><tr><td>71</td><td>SD71</td><td>smoke sensor saloon 2</td><td>33</td></tr><tr><td>72</td><td>SD72</td><td>smoke sensor saloon 1</td><td>33</td></tr><tr><td>73</td><td>SD73</td><td>smoke sensor electronic room</td><td>No cam</td></tr><tr><td>74</td><td>SD74</td><td>smoke sensor driver cab</td><td>No cam</td></tr></table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>N/A</td></tr>
<tr><td>2F_13.Zefiro-Europe.CONCEPT.108</td><td>The PIS MMI in the train staff compartment shall queue the indication of the TCMS and visualize the image of the camera indicated by the TCMS</td><td>Derived Requirement</td><td>4S_10.02.00.-.Passenger Information and Entertainment System - PIES</td><td></td><td>N/A</td></tr>
<tr><td>2F_13.Zefiro-Europe.CONCEPT.107</td><td>In case of activation of the passenger emergency intercom the MMI shall visualize the image of the nearest camera to the location of the relevant intercom device.</td><td>Derived Requirement</td><td>4S_10.02.00.-.Passenger Information and Entertainment System - PIES</td><td></td><td>N/A</td></tr>
</table>

3.2 Events

3.2.1 Function Design

All the system diagnostic shall be sent to the TCMS thorough Ethernet.
Refer to RD4~~RD4~~ for the complete list signal.

3.2.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.224

Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_13.Zefiro-Europe.CONCEPT.76	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_13.Zefiro-Europe.CONCEPT.28			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_13.Zefiro-Europe.CONCEPT.29			

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_13.Zefiro-Europe.CONCEPT.30			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_13.Zefiro-Europe.CONCEPT.31			

4.2 Bus Interface

Supplier to TCMS SW:
TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_13.Zefiro-Europe.CONCEPT.32	RearViewDmiON	L	2F_13.Zefiro-Europe.CONCEPT.60
2F_13.Zefiro-Europe.CONCEPT.110	EnabledCab	L	2F_13.Zefiro-Europe.CONCEPT.60
2F_13.Zefiro-Europe.CONCEPT.33	Camera_M1vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.113	Camera_M1comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.114	Camera_M1comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.115	Camera_T2vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.118	Camera_T2vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.116	Camera_T2comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.117	Camera_T2comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65

Nr	Functional name	Type	Linked to requirement
2F_13.Zefiro-Europe.CONCEPT.119	Camera_M3vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.120	Camera_M3vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.121	Camera_M3comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.122	Camera_M3comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.123	Camera_T4vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.124	Camera_T4vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.125	Camera_T4comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.126	Camera_T4comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.127	Camera_T5vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.128	Camera_T5vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.129	Camera_T5comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.130	Camera_T5comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.131	Camera_M6vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.132	Camera_M6vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.133	Camera_M6comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.134	Camera_M6comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.135	Camera_T7vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.136	Camera_T7vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.137	Camera_T7comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.138	Camera_T7comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.139	Camera_M8vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65
2F_13.Zefiro-Europe.CONCEPT.140	Camera_M81comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65

Nr	Functional name	Type	Linked to requirement
2F_13.Zefiro-Europe.CONCEPT.141	Camera_M8comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65

5 References

RD	Code	Title	Source	Revision
RD1	4pd-pr000086-e	TECHNICAL/COMMERCIAL PROPOSAL Passenger Information System &Video Surveillance System For ETR1000	Selex	4.0
RD2	4pd-pa000010-e	Telecommunication System on board system schematic diagrams	Selex	1.9
RD3	4pd-pa000009-e	Interface drawing / dimension drawing	Selex	1.4
RD4	3EST000225-7439	Interface Control Document (ICD): PIS CCTV V300ZERIFO	BT	C
RD5	EC01P009000B	PIS SFRS	AB	02

6 Annex: **Relations between smoke sensors and cameras**

X		CCctvActiveCamReq	Active CCTV camera request	UINT8	0 = No cam selected 1 = Cam DM1 Vestibule 1 2 = spare 3 = Cam DM1 Comp 1 4 = Cam DM1 Comp 2 5 = Cam TT2 Vestibule 1 6 = Cam TT2 Vestibule 2 7 = Cam TT2 Comp 1 8 = Cam TT2 Comp 2 9 = Cam M3 Vestibule 1 10 = Cam M3 Vestibule 2 11 = Cam M3 Comp 1 12 = Cam M3 Comp 2 13 = spare 14 = Cam T4 Vestibule 1 15 = Cam T4 Vestibule 2 16 = Cam T4 Comp 1 17 = Cam T4 Comp 2

					18 = Cam T5 Vestibule 1 19 = Cam T5 Vestibule 2 20 = Cam T5 Comp 1 21 = Cam T5 Comp 2 22 = Cam M6 Vestibule 1 23 = Cam M6 Vestibule 2 24 = Cam M6 Comp 1 25 = Cam M6 Comp 2 26 = Cam TT7 Vestibule 1 27 = Cam TT7 Vestibule 2 28 = Cam TT7 Comp 1 29 = Cam TT7 Comp 2 30 = Cam DM8 Vestibule 1 31 = spare 32 = Cam DM8 Comp 1 33 = Cam DM8 Comp 2